

MOMIN ABDURREHMAN

AI & Automation Engineer | Data Science | RAG Systems | n8n | API's / LLM Integration

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CGPA: 3.53/4.00

SUMMARY

Data Science student at NUST with hands-on experience delivering production AI systems, ML pipelines, and data analysis projects. Built RAG-based applications and LLM-powered automation engines for paying clients; trained stacked ML ensembles on real-world datasets. Seeking internships in AI Engineering, Data Science/Analytics, or Machine Learning.

TECHNICAL SKILLS

AI & Machine Learning: TensorFlow, PyTorch, Scikit-Learn, OpenAI API, Claude Code, Time-Series Forecasting, Feature Engineering, Model Evaluation, MCP (Model Context Protocol)

AI / LLM: RAG Pipelines, LangChain, LangGraph, Claude API, OpenAI API, Google Gemini, Qdrant, ChromaDB, Vector Embeddings, Prompt Engineering

Automation: n8n, Apify Actors, Clay, Workflow Orchestration, Webhook Pipelines, Batch Processing, API Integration

Backend / APIs: FastAPI, Python, REST APIs, Firebase, MySQL

Frontend: React, React Native, JavaScript

Languages: Python, C++, Java, SQL, JavaScript

EXPERIENCE

Software Developer Intern (Singapore) · Merchaint · Remote

Feb 2026 – Mar 2026

- Delivered two end-to-end AI/automation products during the internship: a RAG-based MCQ exam generator (APE) and an n8n-powered B2B lead generation engine — both deployed to production.
- Worked directly with client requirements to design retrieval architecture, validation logic, and workflow automation systems from scratch in a fast-paced remote environment.
- Gained hands-on exposure to the full AI product lifecycle: scoping, chunking strategy, LLM integration, multi-layer output validation, and deployment on Railway.

Revelo (for Anthropic), Columbia · AI Data Annotator / Code Reviewer · Remote

February 2026

- Performed structured PR-style code reviews on multi-iteration responses generated by competing Claude model variants, evaluating logic correctness, modularity, maintainability, and adherence to software engineering best practices.

PROJECTS

AI / RAG Projects

Multi-Agent Business Intelligence — Python · LangGraph · Gemini 2.0 · Tavily · Chainlit · MCP

- Production-grade multi-agent system for deep company research; four specialised agents (Clarity, Research, Validator, Synthesis) orchestrated in a stateful LangGraph pipeline with conditional routing, human-in-the-loop interrupts, and up-to-3 adaptive retries on insufficient findings.
- Validator-gated quality control with adversarial judge prompt rejects off-topic results; Research Agent generates context-aware Tavily MCP search queries and rotates angles on retry, while SQLite checkpointing persists multi-turn conversation state across restarts.

<https://github.com/Momin-Abdurrehman/multi-agent-business-intelligence>

APE (AI Paper Engine) — React · FastAPI · Qdrant · Claude · OpenAI · Railway

- (Built for a Client) - RAG-based MCQ exam generator deployed for a paying client — question-level PDF chunking ensures the LLM retrieves full questions as context, not arbitrary windows.

- Two-stage validation (programmatic fraction parsing + LLM self-solve pass) guarantees unique, syllabus-aligned answers; per-question regeneration with natural-language feedback.

AI Study Assistant (RAG Bot) — *Python · LangChain · OpenAI · ChromaDB*

- PDF-ingesting Q&A bot with semantic search via ChromaDB vector embeddings; cited answers and accurate cross-document retrieval — prototype that directly led to the APE client engagement.

Automation Projects

AI-Powered B2B Lead Generation Engine — *n8n · Apify · Clay · LLMs · REST APIs*

- (Built for a Client) - Daily Apify scraping of Jobstreet, Indeed, and LinkedIn (SG + MY) feeding an n8n collector–processor pipeline with LLM-scored lead ranking (0–5), deduplication, and automated HTML digest emails.

Machine Learning / Data Analysis Projects

Predictive Maintenance — Turbofan RUL Estimation — *NASA C-MAPSS · XGBoost · LSTM · Streamlit*

- Predicted Remaining Useful Life of jet engines from run-to-failure time-series data using a weighted ensemble of XGBoost and deep learning models (CNN-LSTM, BiLSTM + attention), with KMeans-based condition clustering for feature engineering, unit-wise GroupKFold evaluation (RMSE, MAE, NASA asymmetric score), and an interactive Streamlit dashboard.

<https://github.com/Momin-Abdurrehman/turbofan-rul-prediction>

Pakistan Real Estate Price Prediction — *Python · XGBoost · Scikit-learn · Streamlit*

- 29,220 Zameen.com listings scraped and cleaned to 14,255 records across 6 cities; 22 engineered features including geo signals and Bayesian target encoding for 2,500+ locations.
- Stacked ensemble (RF + GB + XGBoost → Ridge): $R^2 = 0.92$ train/test, 0.88 holdout, 71% of predictions within $\pm 25\%$ — shipped with an interactive Streamlit analytics dashboard.

<https://github.com/Momin-Abdurrehman/pakistan-property-analysis>

T20I Match Winner Prediction — *Python · Scikit-learn · Streamlit · Jupyter*

- 18 years of T20I data (2005–2024); 101 features (ELO, rolling form, H2H, squad) → 40 via mutual information. Best: Logistic Regression at 64.3% accuracy / 0.715 AUC, with a Streamlit demo app.

<https://github.com/Momin-Abdurrehman/T20I-Match-Prediction>

Backend / Database Projects

NUST Sports Management System — *MySQL · Relational Algebra · Database Design*

- 3NF-normalized MySQL schema with 30+ entities (User, Membership, Equipment, Payments) and ACID-compliant transactions; digitized operations for 147 facilities serving up to 3,000 daily users.
- Complex SQL queries and Relational Algebra expressions to automate membership billing, equipment maintenance workflows, and real-time attendance analytics; composite indexing on high-traffic fields for query performance.

Daily Zikr — Islamic Adhkar App — *Flutter · Dart · Firebase · SharedPreferences*

- Cross-platform app (Android, iOS, Web) from a single Flutter codebase - 38 authenticated supplications, 120 rotating hadiths, streak tracking, Hijri calendar, and drag-and-drop reordering.

<https://daily-zikr-app.web.app/>

EDUCATION

Bachelor of Science in Data Science | NUST — National University of Sciences & Technology, Islamabad | 2024 – Present

Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Systems, Linear Algebra, Advance Statistics, Multi-Variable Calculus, Artificial Intelligence, Computer Networks, Computer Organization and Assembly Language.